Mini Intralogistics Simulator: Heterogeneous Fleet Management – Documentation

1. Introduction

This project demonstrates a discrete-event simulation (DES) framework for modeling aerospace intralogistics systems with heterogeneous resources under both nominal and exceptional production flows. The simulator focuses on line-side logistics, warehouse-to-line picking, kitting, supermarkets, and returns management. It provides insights on fleet management, resource allocation, and resilience under uncertainty.

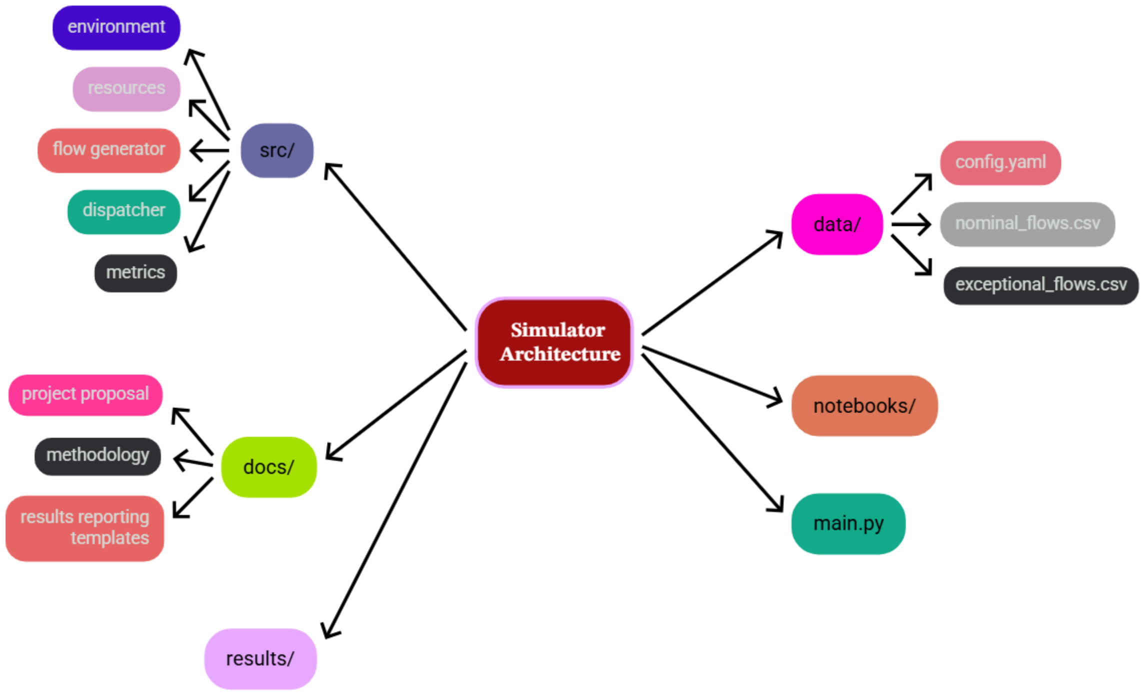
2. Project Objectives

The main objectives of the project are:
1. Simulate nominal production flows and scheduled requests.
2. Incorporate exceptional flows such as replanning, quality issues, rework, missing items, and returns.
3. Model heterogeneous resources including automated AGVs, semi-automated conveyors, and human operators.
4. Evaluate dispatching policies using rule-based and priority management approaches.
5. Analyze performance metrics such as service robustness, capacity utilization, and resilience indicators.

3. Simulator Architecture

The project is structured into multiple modules:
- src/: Simulation source code including environment, resources, flow generator, dispatcher, and metrics.
- data/: Configuration and test data including config.yaml, nominal_flows.csv, and exceptional_flows.csv.
- docs/: Documentation for project proposal, methodology, and results reporting templates.
- notebooks/: Jupyter notebooks for simulation analysis.
- results/: Output and metrics generated by the simulator.

- main.py: Entry point for running the simulation.



4. Key Features

- Nominal Flows: Handle standard scheduled production requests.
- Exceptional Flows: Manage late replanning, quality events, rework, missing items, and returns.
- Heterogeneous Resources: Integrate AGVs, conveyors, and human operators.
- Dispatching Policies: Implement rule-based and priority-based management.
- Performance Metrics: Measure service robustness, capacity utilization, and resilience indicators.

5. Methodology

The methodology involves creating a discrete-event simulation model using SimPy in Python. Nominal and exceptional flows are generated to mimic real production scenarios. Heterogeneous resources are modeled with attributes such as processing times, availability, and operational constraints. Dispatching policies determine the order in which tasks are assigned to resources. Performance metrics are collected and analyzed for resilience and efficiency evaluation.

6. Practical Implementation

The simulation environment can be configured using parameters in config.yaml. Users can define the number of orders, machine processing times, and available capacities. By running main.py, the simulator executes the DES model and outputs performance reports, batch grouping, and scheduling results.

7. Research Questions

1. How do nominal and exceptional flows interact in line-side logistics?
2. What dispatching policies minimize shortages and service delays?
3. How much capacity slack is needed for resilience under uncertainty?
4. How do heterogeneous resources compare in handling mixed workloads?

8. Expected Outputs

- Discrete-event simulation model for aerospace intralogistics.
- Comparative analysis of 3-4 dispatching policies.
- Performance metrics under different scenarios.
- Recommendations for fleet dimensioning and resource planning.
- Academic publication-ready results.

9. Simulation Results and Visual Analysis

The simulation results were converted into graphical visualizations to support interpretation of system performance, flow behavior, resource usage, and policy comparison. These visuals help explain how the intralogistics system behaves under nominal and exceptional flows.

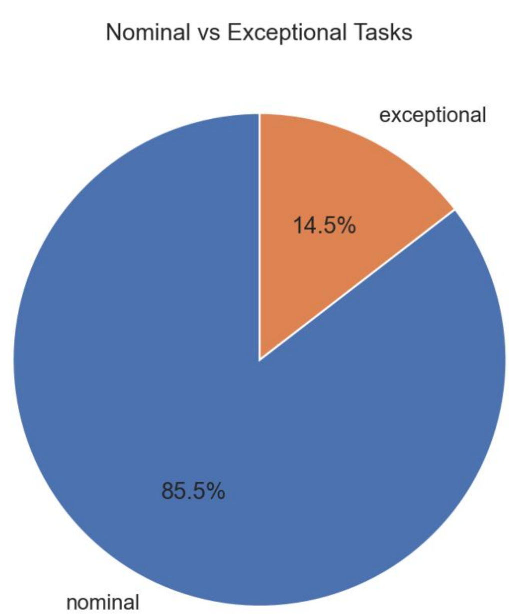


Figure 1: Flow Mix Analysis

This figure shows the proportion of nominal and exceptional tasks in the simulation. Nominal tasks represent regular planned intralogistics operations, while exceptional tasks represent urgent, unexpected, or disruptive flows. The chart helps identify how much of the workload is routine and how much is caused by uncertainty. This is important for evaluating the resilience of the system under mixed operating conditions.

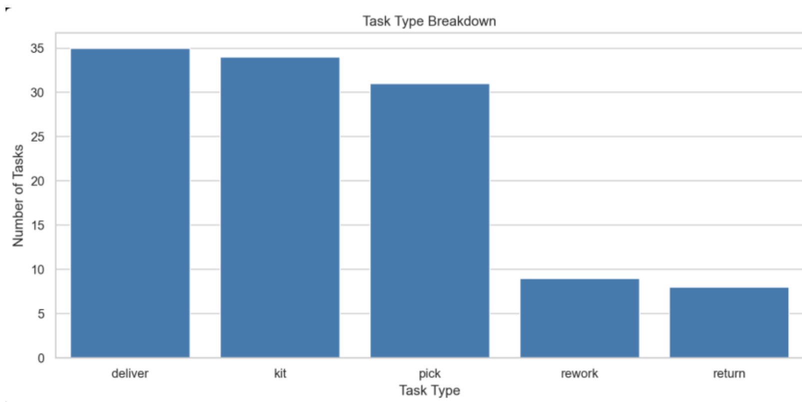


Figure 2: Task Type Breakdown

This figure presents the distribution of different task types such as delivery, picking, kitting, returns, and rework. It shows which operational activities appear most frequently in the simulated environment. A higher value indicates that the corresponding task type has a

stronger impact on system workload. This visualization helps identify the main sources of logistics demand in the aerospace assembly context.

Figure 3: Resource Allocation

This figure shows how tasks are assigned to different heterogeneous resources such as AGVs, conveyors, and human operators. It helps evaluate whether the workload is balanced across the fleet or concentrated on a specific resource. If one resource receives most of the tasks, it may indicate a dependency or potential bottleneck. This chart supports the analysis of fleet utilization and dispatching effectiveness.

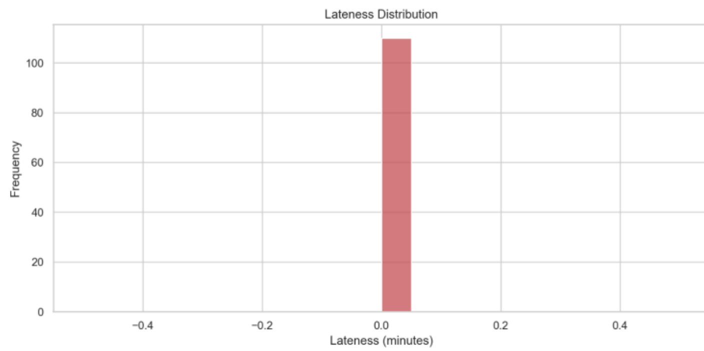


Figure 4: Lateness Distribution

This figure shows the distribution of task lateness in minutes. It is used to evaluate whether tasks are completed on time or delayed. A concentration of values near zero indicates strong service performance and reliable task completion. This visualization supports the assessment of service level, robustness, and resilience under uncertainty.

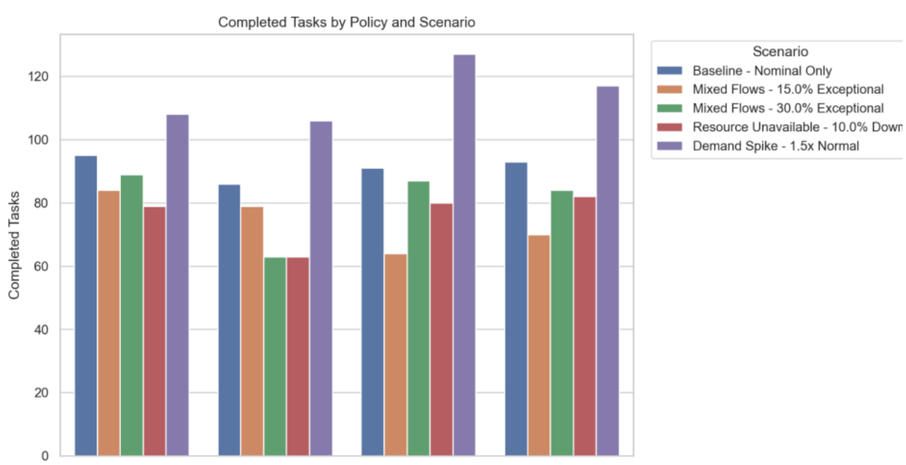


Figure 5: Dispatching Policy Comparison

This figure compares the number of completed tasks across different dispatching policies and scenarios. It allows comparison between FIFO, priority-based, load-balancing, and preemptive strategies. Higher bars indicate that a policy completed more tasks under a given scenario. This visualization helps identify which dispatching approach performs better for heterogeneous fleet management.

10. Project Timeline

Week 1: Environment setup and nominal flow modeling.
Week 2: Implement exceptional flows, resource models, and basic simulation.
Week 3: Develop and test dispatching policies.
Week 4: Conduct experiments, analyze results, optimize, and finalize documentation.

11. Contact

For questions or improvements, refer to the project proposal document in the docs/ folder or contact the project author. aenaazhar1@gmail.com